

ADI SHANKARA INSTITUTE OF ENGINEERING & TECHNOLOGY



(An ISO 9001:2008 Certified Institution)
KALADY, KERALA



DEPARTMENT OF ELECTRICAL & ELECTRONICS
ENGINEERING



SEPTEMBER 2015

A

Seminar Report on

MULTIPHASE PWM CONTROL STRATEGY FOR A PMSM WITH SEGMENTED STATOR WINDINGS

SUBMITTED BY

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CERTIFICATE

*Certified that this is the Bonafide Record of the Seminar entitled
“MULTIPHASE PWM CONTROL STRATEGY FOR A PMSM
WITH SEGMENTED STATOR WINDINGS” submitted by
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*during the year 2015 in partial fulfillment for the award of
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ACKNOWLEDGEMENT

I consider myself proud to be a part of Adi Shankara Institute of Engineering and Technology, Kalady, the institution that stood by me in all my endeavors, which provided me with an opportunity to carry out this seminar presentation. I wish to thank our Director, **Dr S.G. Iyer**, our Principal, **Dr. Neelakantan P.C.** and our HOD of the Electrical and Electronics Department, **Mrs. Krishnakumari T.** for their kind support and encouragement in carrying out this seminar presentation successfully. Most importantly, I would like to thank my internal guide **Mrs. Hima T.** for her guidance, inspiration and for answering all my queries satisfactorily for the seminar presentation. I also thank the project coordinators whose constant persistence and support helped in the completion of presentation. Finally, I would like to thank my parents, teachers and friends who in some way or other helped me in successful completion of this work.

ABSTRACT

A multiphase converter topology with an innovative PWM strategy for the control of a PM synchronous motor with segmented stator windings to operate as an electronic geared motor is proposed. Compared with conventional three- phase inverter topology, the proposed converter topology with associated PWM modulation strategy has advantages of reduced conduction loss, improved efficiency under light load condition, and expanded torque-speed operating ranges with a same dc-link voltage. The interleaved PWM strategy provides advantages of ripple current reduction and conduction losses reduction.

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LIST OF ABBREVIATIONS

Electronically Controlled Motor (ECM)

Permanent Magnet Synchronous Motor (PMSM)

Space-Vector-Modulation (SVM)

Pulse Width Modulation (PWM)

Electric vehicles (EV)

Electromagnetic Interference (EMI)

CHAPTER 1

INTRODUCTION

Recent years have seen a growing need for large diameter direct drives, particularly in precision industry applications. Such drives are required to provide a variety of operating conditions and customers demand low-cost tailor-made drive solutions.

Segmented electric machine have been gaining a lot of interest over last few years on account of high flexible and low manufacturing costs. The modular concept based on rotor segments and stator segments which can share a flexible production line for a wide range of applications. This provides possibility for manufacturing automation of small electric motors with low cost and flexible applications. When a motor is designed with a high number of pole pairs, the modular design concept that comprises a plurality of same motor segments has advantages of automatic manufacturing and flexible applications. Such segmented motors which can be assembled with stator or rotor segments are characterized in that they are highly flexible when it comes to control and drive aspects. Various converter topologies and PWM control schemes have been developed for ac motors with three-phase winding.

A PWM control scheme for a motor with multiphase open-end winding topology, it provides an independent PWM control for each winding and also provides multi-level voltage conversion with separate dc voltage sources. An electric machine with open-end windings provides flexibility for power converters to control the corresponding winding as an independent load.

Multiphase drives have various applications including electric ship propulsion, locomotive traction, electric and hybrid electric vehicles, and high-power industrial applications. As a consequence, there has been a substantial increase in the interest for such drive systems worldwide. The reasons for potential use of multiphase rather than three-phase drives are the availability of various control schemes and use of additional degrees of freedom that exist in multiphase machines.

CHAPTER 2

SEGMENTED ELECTRIC MACHINE

When a motor is designed with a high number of pole pairs, the modular design concept that comprises a plurality of same motor segments has advantages of automatic manufacturing and flexible applications. Such segmented motors which can be assembled with stator or rotor segments are characterized in that they are highly flexible when it comes to control and drive aspects. Various converter topologies and PWM control schemes have been developed for ac motors with three-phase windings

A PWM control scheme for a motor with multiphase open-end winding topology, provides an independent PWM control for each winding and also provides multi-level voltage conversion with separate dc voltage sources. An electric machine with open-end windings provides flexibility for power converters to control the corresponding winding as an independent load. Space vector PWM has been applied to a five-phase PMSM with open-end windings by using five H-bridge VSIs with a common dc source .A three-phase PMSM with open-end windings can be controlled with three single-phase H-bridge inverters with a common dc bus . In contrast to the disadvantage of doubling the power semiconductor devices, this topology provides advantages such as minimization of switching losses, balancing of switching rate between the three H-bridges, insensitivity to duty cycle, maximization of the drive performance, and reduction of the zero- sequence current ripple. A segmented motor with modular segmented stator windings are designed and manufactured with identical segments and are assembled to be fixed on the laminated stator core.

An electronically controlled motor (ECM) is a motor designed with integrated control electronics. A segmented ECM can provide design flexibility with the control and drive of these stator windings by using innovative converter topology and PWM strategy. To explore advantages of PWM control of segmented motors with open-windings, this paper proposes a multiphase PWM control scheme for a series of H-bridge inverters with a common DC link voltage in applications to a three-phase PMSM with 8-pole and 12-slot segmented stator windings. Compared with the conventional three-phase inverter drive with a common neutral, this multi-phase H-bridge inverter drive can provide a synchronized and coordinated PWM control of each winding, and therefore, these windings are controlled independently while are functionally be configured either in series or parallel. A series connection of the stator windings in same phase is equivalent to an increase of the motor back emf constant, while a parallel connection is equivalent to an increase of the motor torque constant. This multi-phase converter topology with a coordinated PWM strategy illustrates the concept of electronic geared motor control. Compared with the conventional three-phase inverter motor drive, the proposed multiphase inverter for a segmented motor can achieve wider speed and torque control range with a same DC-link voltage, while at the same time improve the efficiency over a wide speed control range.

CHAPTER 3

MULTI-PHASE CONVERTER FOR SEGMENTED PMSM

Switched stator windings method has been developed to change the torque-speed characteristics of a PMSM by adjusting the connections of the stator windings with open-ends. However, this mechanical connection must be done before its application and can not be changed during its operation. The parallel connection of the same phase windings will also result current unbalancing problems. On the other hand, multiphase inverter can control the current of each segmented windings to achieve accurate current sharing under parameter uncertainties. PWM interleaving technique can also be applied for the control of a multiphase inverter for the segmented ECM. A three-phase interleaved PWM strategy, as shown in Fig. 3.1,

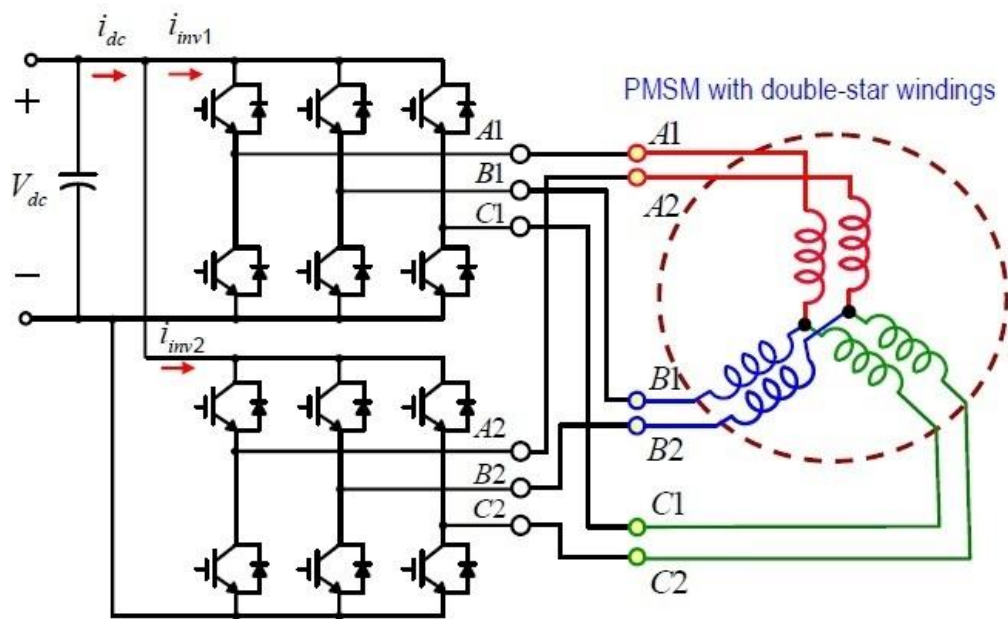


Fig. 3.1. Schematic of a three-phase interleaved inverter motor drive with double-star windings.

has been employed for the control of two three-phase PWM inverters of a double- star connected segmented motor to reduce the ripple currents and thus the size of the dc-link capacitor in applications to EV.

Interleaved PWM technique can be applied to converters with fixed switching frequency, the phase shift of the PWM signals has an effect to reduce the current ripples for its summing current. A segmented motor is a motor can be assembled with identical stator or rotor segments. Fig. 3.2

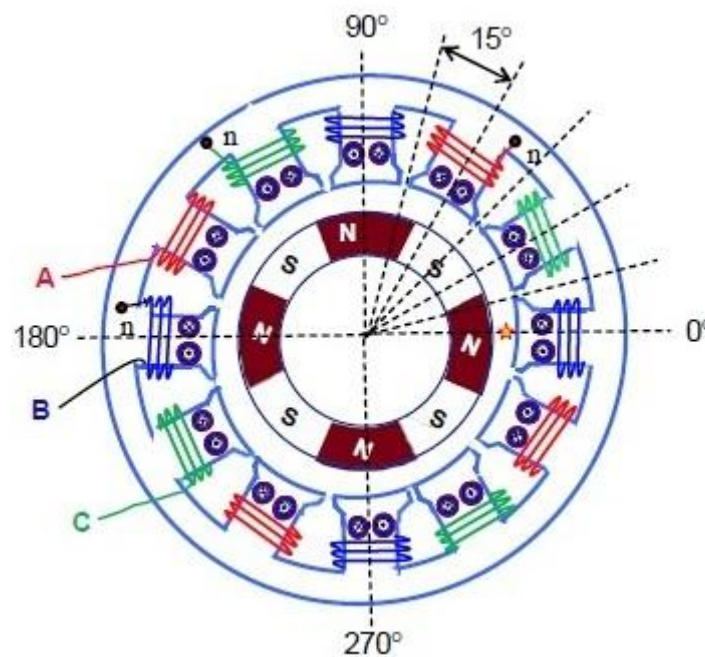


Fig.3.2. Segmented PMSM with open-end stator windings. Cut-view of a 12-slot, 8-pole segmented PM synchronous motor.

shows the structure of a surface mounted PM synchronous motor with segmented stator windings, the rotor of motor has eight poles and the stator has twelve slots. Each phase has four independent segmented coils which are open-end winding as shown in Fig. 3.3.

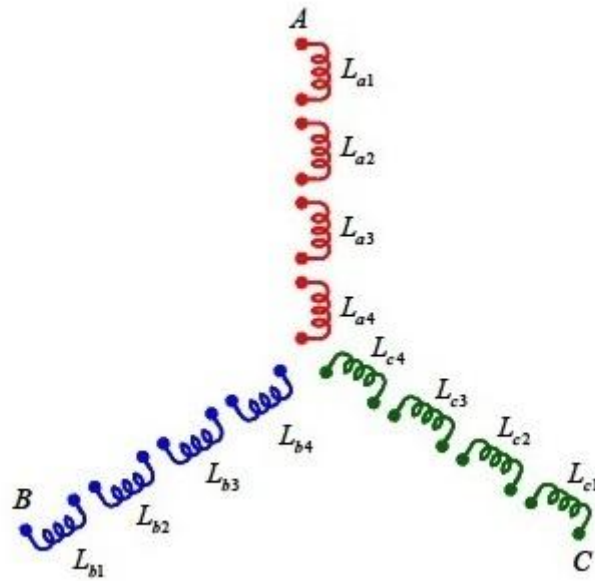


Fig. 3.3. Segmented PMSM with open-end stator windings. Illustrative schematics of the segmented stator with open-end windings.

The ends of these segmented windings are distributed around the stator and can be either connected with wire bonds or a PCB mounted on the top of these segments. Various converter topologies with 4-quadrant operation can be used to control the current of these windings to generate a rotating magnetic field for ac motor control. A full-bridge converter has advantages of inherent symmetry, full dc-link voltage output, 4-quadrant operation, can be either operating in voltage mode or current mode control, and therefore is selected to realize the proposed multiphase converter for segmented ECM.

Fig.3.4 shows the multiphase converter topology for the proposed multiphase PWM control scheme. Each segment winding can be viewed as a controllable stator magnetic field located on its designated circumferential position.

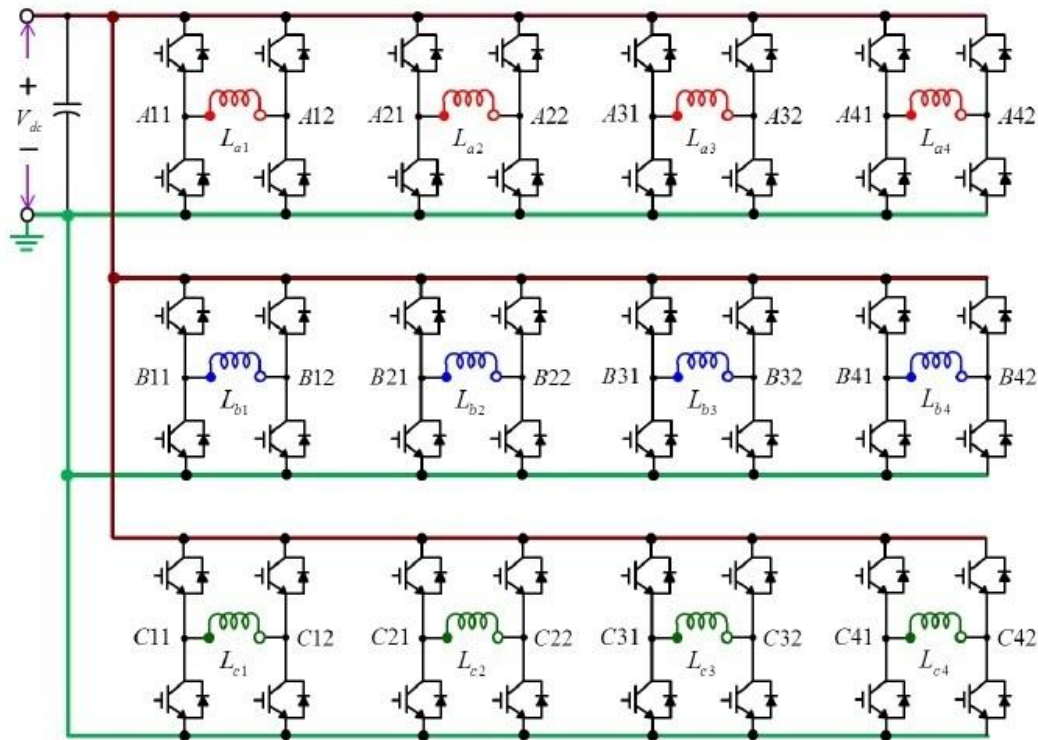


Fig. 3.4. Proposed multiphase converter topology.

The vector summation of these segmented fields will produce a composite stator magnetic field. The interaction of this stator magnetic field and rotor magnetic field will produce a controllable electric torque. Thus, a phase winding with N multiple segments provides $2N-1$ degree- of-freedom for the control of the stator field of the corresponding phase. There are four segments for each phase of the three- phase segmented PMSM, therefore, a total of 12 windings need to be controlled. Fig. 3.4 shows there are twelve H-bridge converters to control the voltage applied to each winding of the segmented motor. This multiphase converter requires a high number of power semiconductor device, $4 \times 3 \times N$ power switches, for an N - winding per phase segmented motor. This multiphase converter provides control flexibility while at the same time more involved control circuitry and sophisticated timing control are required. In spite of the more complicated circuitry, there are many benefits of this multiphase converter. The converter provides advantages of distributed power conversion to the segmented motor. Each segment can be viewed as a single-phase motor and each motor contributes its generated torque to the whole motor. The voltage applied to the motor are shared by more power devices, therefore, lower rating of power devices can be employed. The current ratings of these power devices are kept the same.

The dc-link voltage can be reduced to

$$V_{dc} = \frac{V_{dc,3w}}{N} \text{ -equation(1)}$$

where N is the number of segments per phase and $V_{dc,3w}$ is the dc-link voltage required by the original three-wire inverter.

Another benefit of the multiphase converter is distributed power dissipation. In the conventional six- pack power module for the three-phase inverter, power losses are concentrated, while for the multiphase converter the losses are distributed in N H-bridge power module and loss for each module is also much smaller. This provides great advantage for the thermal design of high power ECM. The multiphase converter also provides lower electromagnetic interference (EMI). Each H-bridge converter can be placed very close to the segment winding, this significantly reduce the stray inductance and noises coupling of the power circuit. The common- mode EMI is also significant reduced due to the much lower dc-link voltage. Interleaved PWM techniques can be applied to multiphase converter to reduce the composite torque ripples of the segmented motor. Symmetric or asymmetric interleaving techniques can be applied to the segmented drive to reduce EMI.

CHAPTER 4

MULTIPHASE PWM CONTROL STRATEGY

The interleaved PWM for the multiphase converter can also reduce the torque ripples through the reduction of composite torque in the same phase. A multiphase PWM control strategy, as shown in Fig. 4.1,

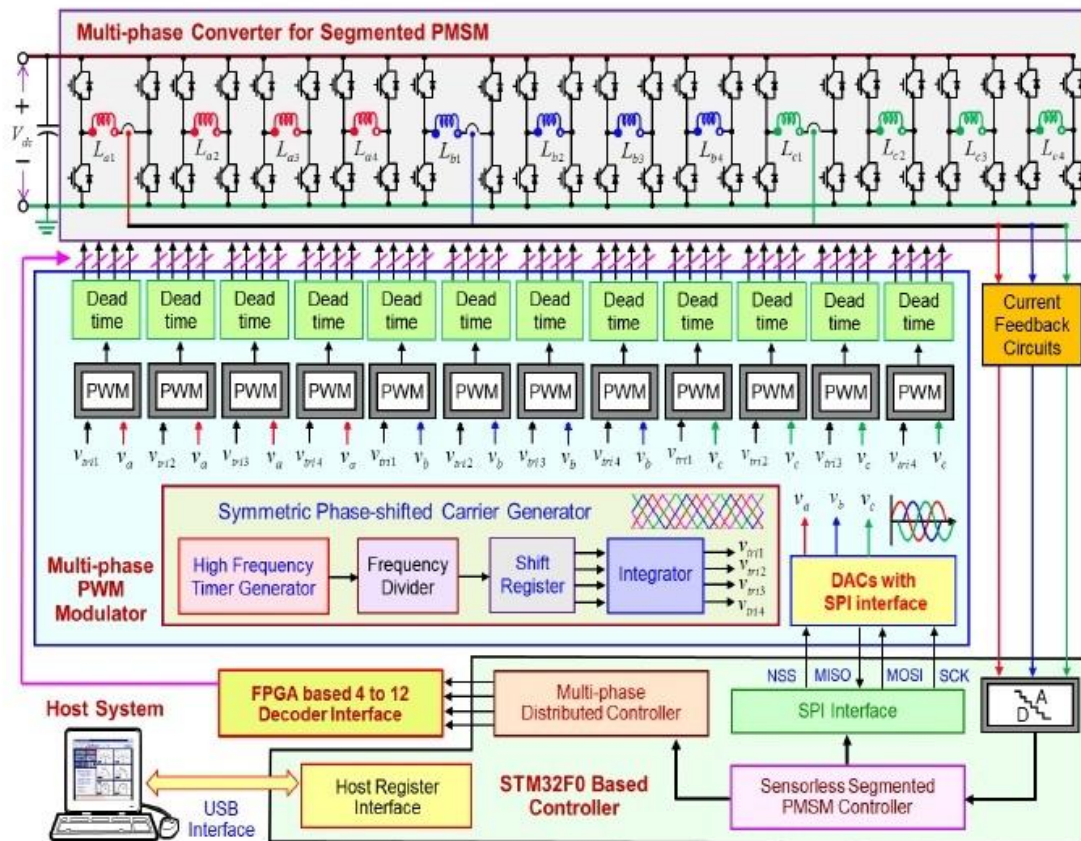


Fig. 4.1. Block diagram of the multiphase PWM control strategy.

is developed for the reduction of the torque ripples by phase shifting control and to reduce the converter losses by phase shedding control. Therefore, the multiphase converter can improve the overall speed-torque control range, efficiency, EMI, and thermal design at the cost of more complicated control and power circuitry.

4.1. Phase Shedding Control

In consideration of the magnetic field symmetry for smooth operation, the segmented PMSM with 12-slot, 8- pole is defined with operating modes as shown in Fig. 4.2.

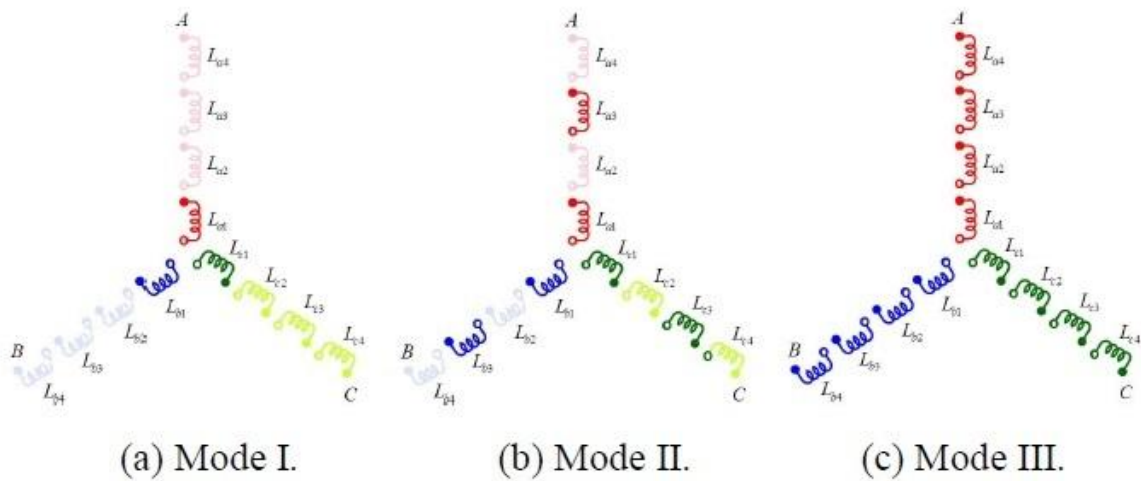


Fig. 4.2. Operating modes of the multiphase PWM strategy.

One, two, and four active windings are selected for operating modes I, II, and III, respectively. The light- colored coils of each phase, as shown in Fig. 4.2, are disabled in the corresponding operating modes. The developed electric torque of the segmented motor is a summation of torques produced by excited windings and can be expressed as

$$T_e = \sum_{i=1}^N \tau_{ai} + \tau_{bi} + \tau_{ci} \text{ -equation (2)}$$

where N is the number of the coil per phase, i is number of the segmented winding in its corresponding phase, τ_{ai} represents the produced torque of a-phase by the i-th winding.

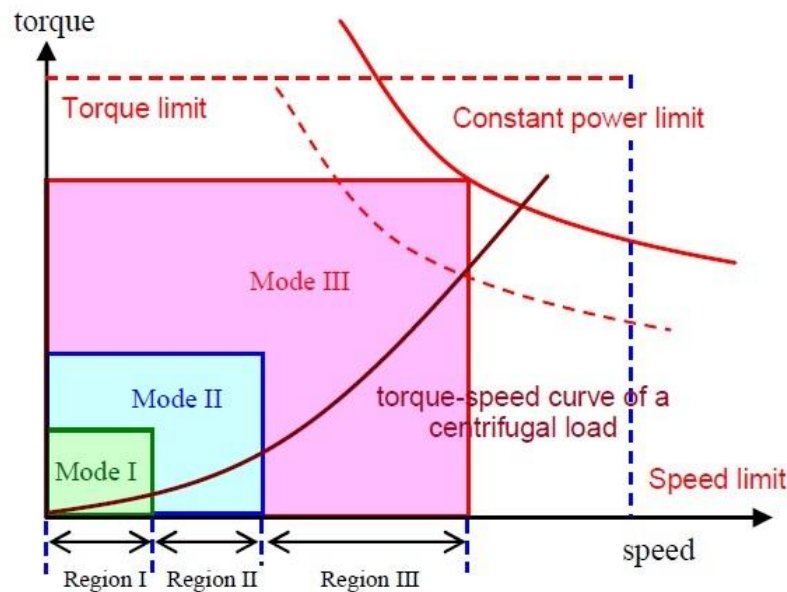


Fig. 4.3. Torque-speed region for different operating modes.

If the phase current is sinusoidal and be kept in phase with the back emf, then the summated torque will be a constant. Fig.4.3 shows the torque-speed operating region for the defined operating modes of the multiphase converter. When more segmented windings are activated, higher voltage and current can be applied to the motor, and thus increase the motor output power. The maximum rated power is limited by the motor due to its thermal dissipation capability. The maximum speed limit is caused by many factors, such as load characteristics, available dc-link voltage, field-weakening control, and motor itself. The maximum torque limit is usually limited by the available maximum phase current. A motor is designed with a given torque-speed operation region for both continuous and intermittent operation. When the motor is applied to a specific load, its operating point is determined by the command and the load torque-speed characteristics. Fig. 4.3 also shows the torque-speed curve of a centrifugal load. It can be observed that when the operation speed is changed, its operating point may operate in different operating modes. This can be an advantage for energy saving for light load operation. It can be observed that if the motor is running at low speed and operating in region I for mode I operation, then only one of the four bridge converters for each phase are activated. This has a significant effect for efficiency enhancement for light load operation.

4.2. Phase Shift Control

When multiple PWM converters are connected in parallel, it is advantageous to operate all converters at the same frequency but with certain phase shift among their PWM signals. The technique is referred as interleaving technique and has been widely used in high current DC-DC converters. The interleaving technique provides major advantages for current sharing and current ripple reduction. This technique can also be applied to a multiphase inverter for current sharing and current ripple reduction.

The phase shift of the i -th carrier of an N -phase interleaved PWM converter is

$$\theta_{shift}(i) = \frac{2(i-1)*180^\circ}{N} \quad \text{-equation(2)}$$

where N is the number of the modules one phase.

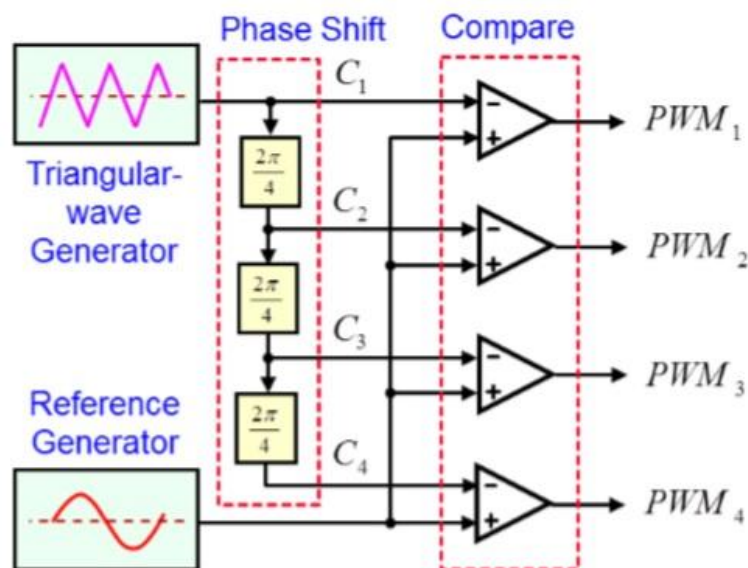


Fig. 4.4. Symmetric interleaving phase shifted carrier for one phase with four segmented coils

Fig. 4.4 shows the block diagram of symmetric interleaving phase shifted carrier generator for one phase with four segmented coils. Because the segmented windings are independently controlled, and their phase currents are not summated, therefore the current ripple reduction effect will not appear. However, because the instantaneous torque is proportional to the phase current, the output torque ripples can be reduced due to the interleaving PWM control. Moreover, the input current ripple can also be reduced by the interleaved PWM control. This ripple cancellation effect can not only reduce input filtering requirement but also the switch frequency for decreasing the switching loss.

CHAPTER 5

CONCLUSION

A multiphase PWM control strategy for the PM synchronous motor with segmented stator windings to operate as an electronic geared motor has been proposed. A multiphase PWM modulation strategy is developed for the control of the segmented PMSM to achieve a same torque-speed performance with a lower dc-link voltage at the cost of an increase of switching devices with lower voltage ratings. The interleaved PWM strategy provides advantages of ripple current reduction and conduction losses reduction. Phase shedding technique can be used to improve system efficiency under light load operating condition.

Therefore, this method can be useful in applications to motor drives with requirements of high power quality and high efficiency improvement. The tradeoff of these advantages is an increase of N times power semiconductor devices with $1/N$ voltage rating and same current rating. With the advances of power semiconductor devices and control ICs, the proposed multiphase distributed control scheme will gain its advantages in efficiency improvement and thermal package design for advanced ECM.

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